

# Problem Sheet — IMC Training

May 13, 2026

**Problem 1.** (IMC 2022, Day 1, Problem 2) Let  $n$  be a positive integer. Find all  $n \times n$  real matrices  $A$  with only real eigenvalues satisfying

$$A + A^k = A^T$$

for some integer  $k \geq n$ .

**Problem 2.** (IMC 2021, Day 1, Problem 2) Let  $n$  and  $k$  be fixed positive integers, and let  $a$  be an arbitrary non-negative integer. Choose a random  $k$ -element subset  $X$  of  $\{1, 2, \dots, k + a\}$  uniformly and, independently of  $X$ , choose a random  $n$ -element subset  $Y$  of  $\{1, \dots, k + n + a\}$  uniformly. Prove that the probability

$$\mathbb{P}(\min(Y) > \max(X))$$

does not depend on  $a$ .

**Problem 3.** (IMC 2020, Day 2, Problem 5) Find all twice continuously differentiable functions  $f : \mathbb{R} \rightarrow (0, +\infty)$  satisfying

$$f''(x) f(x) \geq 2(f'(x))^2$$

for all  $x \in \mathbb{R}$ .

**Problem 4.** (IMC 2022, Day 1, Problem 3) Let  $p$  be a prime number. A flea is staying at point 0 of the real line. At each minute, the flea has three possibilities: to stay at its position, or to move by 1 to the left or to the right. After  $p - 1$  minutes, it wants to be at 0 again. Denote by  $f(p)$  the number of its strategies to do this. Find  $f(p)$  modulo  $p$ .

**Problem 5.** (IMC 2020, Day 2, Problem 7) Let  $G$  be a group and  $n \geq 2$  be an integer. Let  $H_1$  and  $H_2$  be two subgroups of  $G$  that satisfy

$$[G : H_1] = [G : H_2] = n \quad \text{and} \quad [G : (H_1 \cap H_2)] = n(n - 1).$$

Prove that  $H_1$  and  $H_2$  are conjugate in  $G$ .

**Problem 6.** (IMC 2021, Day 2, Problem 8) Let  $n$  be a positive integer. At most how many distinct unit vectors can be selected in  $\mathbb{R}^n$  such that from any three of them, at least two are orthogonal?

**Problem 7.** (IMC 2022, Day 2, Problem 7) Let  $A_1, A_2, \dots, A_k$  be  $n \times n$  idempotent complex matrices such that

$$A_i A_j = -A_j A_i \quad \text{for all } i \neq j.$$

Prove that at least one of the given matrices has rank  $\leq \frac{n}{k}$ . (A matrix  $A$  is called idempotent if  $A^2 = A$ .)

**Problem 8.** (IMC 2020, Day 2, Problem 8) Compute

$$\lim_{n \rightarrow \infty} \frac{1}{\log \log n} \sum_{k=1}^n (-1)^k \binom{n}{k} \log k.$$

(Here  $\log$  denotes the natural logarithm.)

**Problem 9.** (IMC 2021, Day 2, Problem 7) Let  $D \subseteq \mathbb{C}$  be an open set containing the closed unit disk  $\{z : |z| \leq 1\}$ . Let  $f : D \rightarrow \mathbb{C}$  be a holomorphic function, and let  $p(z)$  be a monic polynomial. Prove that

$$|f(0)| \leq \max_{|z|=1} |f(z) p(z)|.$$

**Problem 10.** (IMC 2020, Day 1, Problem 3) Let  $d \geq 2$  be an integer. Prove that there exists a constant  $C(d)$  such that the following holds: for any convex polytope  $K \subset \mathbb{R}^d$ , which is symmetric about the origin, and any  $\varepsilon \in (0, 1)$ , there exists a convex polytope  $L \subset \mathbb{R}^d$  with at most  $C(d) \varepsilon^{1-d}$  vertices such that

$$(1 - \varepsilon)K \subseteq L \subseteq K.$$